

2.3-Testing_some_stereotypes

RStudio assignment · Team LAGARDE / THEBUWANA / GILLES

Alexis LAGARDE

Vidusha THEBUWANA

Chloé GILLES

1 October 2026

About this report

- **GitHub repository:** <https://github.com/AlexisLag-student/IEAP-Series02-RStudio>
- **Structure:** this file is the *master document*. Each section is written in its own file in `sections/` and included here at render time, so that each team member edits a different file and we never create merge conflicts.

Section	File	Owner
2.1 What is a statistical test?	<code>sections/_01-concepts.qmd</code>	Alexis LAGARDE
2.2 Effect of treatment over time	<code>sections/_02-prepost.qmd</code>	Vidusha THEBUWANA
2.3 Testing some stereotypes	<code>sections/_03-snore.qmd</code>	Chloé GILLES
3 Epilogue	<code>sections/_04-epilogue.qmd</code>	Alexis LAGARDE
Git workflow, challenges, checklist	<code>sections/_05</code>	Done all together

Statistical tests: what they are and when to use them

What is a median? A mean? The mean is the sum of the values divided by their number. The median is the value that splits the ordered data into two halves (50% below, 50% above). The median is robust to extreme values; the mean is not (Manikandan 2011).

What is a variance? A standard deviation? The variance is the mean of the squared deviations from the mean (divided by $n - 1$ for a sample). The standard deviation (SD) is its square root, so it is expressed in the same unit as the data. The SD describes the spread of the individual values; it should not be confused with the standard error, which describes the precision of the mean (Altman and Bland 2005).

What is a normal distribution? A symmetric, bell-shaped distribution fully described by its mean and SD. About 68% of the values fall within ± 1 SD of the mean and about 95% within ± 1.96 SD (Altman and Bland 1995).

What is a statistical test? A procedure that uses sample data to decide whether the observed result is compatible with a null hypothesis H_0 (e.g. “no difference between groups”). It computes a test statistic and the probability of observing a statistic at least as extreme if H_0 were true (Greenland et al. 2016).

When to use a statistical test? When we want to generalise from a sample to a population: for example, to decide whether a difference between two groups, or between before and after a treatment, is larger than what chance sampling variability alone would produce (Banerjee et al. 2009).

What is a parametric test? A non-parametric test? A parametric test assumes that the data follow a known distribution (usually normal) and compares its parameters (e.g. means for the t -test). A non-parametric test makes no assumption on the shape of the distribution and usually works on the ranks of the data (e.g. Wilcoxon, Mann–Whitney) (Nahm 2016).

What are the assumptions of parametric tests? Independent observations, approximately normal data (or normal differences for paired tests), and for some tests equal variances between groups. Normality can be checked with a Q–Q plot or a Shapiro–Wilk test (Ghasemi and Zahediasl 2012); Welch’s t -test removes the equal-variance assumption (Welch 1947).

What are the assumptions of non-parametric tests? Independent observations (or independent pairs), a variable that is at least ordinal and, to interpret the Mann–Whitney test as a comparison of medians, distributions of similar shape in both groups (hart2001?). The Wilcoxon signed-rank test also assumes that the differences are distributed symmetrically (Wilcoxon 1945).

What is a p-value? The probability, assuming H_0 is true (and all model assumptions hold), of obtaining a result at least as extreme as the one observed. It is *not* the probability that H_0 is true, and it does not measure the size or importance of an effect (Wasserstein and Lazar 2016).

What is the risk of error when using a statistical test? Type I error (α): rejecting H_0 when it is true (false positive), usually fixed at 5%. Type II error (β): failing to reject H_0 when it is false (false negative); $1 - \beta$ is the power of the test. Running many tests inflates the overall type I error (Banerjee et al. 2009).

What is the difference between a paired and an unpaired test? A paired test compares two measurements taken on the *same* individuals (e.g. before and after training): it analyses the within-subject differences. An unpaired test compares two *independent* groups (e.g. snorers vs non-snorers). Using an unpaired test on paired data ignores the within-subject correlation and loses power (Bland and Altman 2009).

2.2 Effect of treatment over time

2.2.1 Looking at the data

Loading the necessary libraries

```
# Load standard libraries
library(tidyverse)
```

Loading the data

```
# Loading the data
data_long <- read_csv("./Data/PrePost.csv")

# Adding Subject IDs (assuming 8 patients in the exact same order)
# and convert 'time' to a factor so 'Before' always plots first
data_long <- data_long %>%
  mutate(ID = rep(1:8, times = 2)) %>%
  mutate(time = factor(time, levels = c("Before", "After")))

# Pivot to wide format to easily calculate the difference
data_wide <- data_long %>%
  pivot_wider(names_from = time, values_from = perf) %>%
  mutate(difference = After - Before)

head(data_wide)
```

```
# A tibble: 6 x 4
      ID Before After difference
  <int> <dbl> <dbl>      <dbl>
1     1    125   127         2
2     2    130   132         2
3     3    132   133         1
4     4    135   136         1
5     5    136   139         3
6     6    138   141         3
```

Variable Descriptions

Variable	Description	Unit	Possible values	Expected change
perf	Physical performance score of the patient	not given in the file	positive values; here 125 to 148	↑ after rehabilitation
time	Moment of the measurement	—	Before, After	— (factor)

2.2.2 The analysis

Assessing normality

First will check the normality by Shapiro-Wilk Normality Test, before deciding using Parametric or non-parametric test.

```
# Check normality assumption
shapiro_test <- shapiro.test(data_wide$difference)
print(shapiro_test)
```

Shapiro-Wilk normality test

```
data: data_wide$difference
W = 0.89896, p-value = 0.2828
```

A Shapiro-Wilk test was conducted on the differences between post- and pre-treatment performance scores to evaluate the assumption of normality. The results show that the distribution of score differences does not significantly deviate from a normal distribution ($W=0.899, p=0.283$). Because $p>0.05$, the assumption of normality is met, validating the choice of a parametric paired t-test for the main analysis.

Running the paired t-test

```
# Run the parametric paired t-test
t_test_result <- t.test(data_wide$After, data_wide$Before, paired = TRUE)
print(t_test_result)
```

Paired t-test

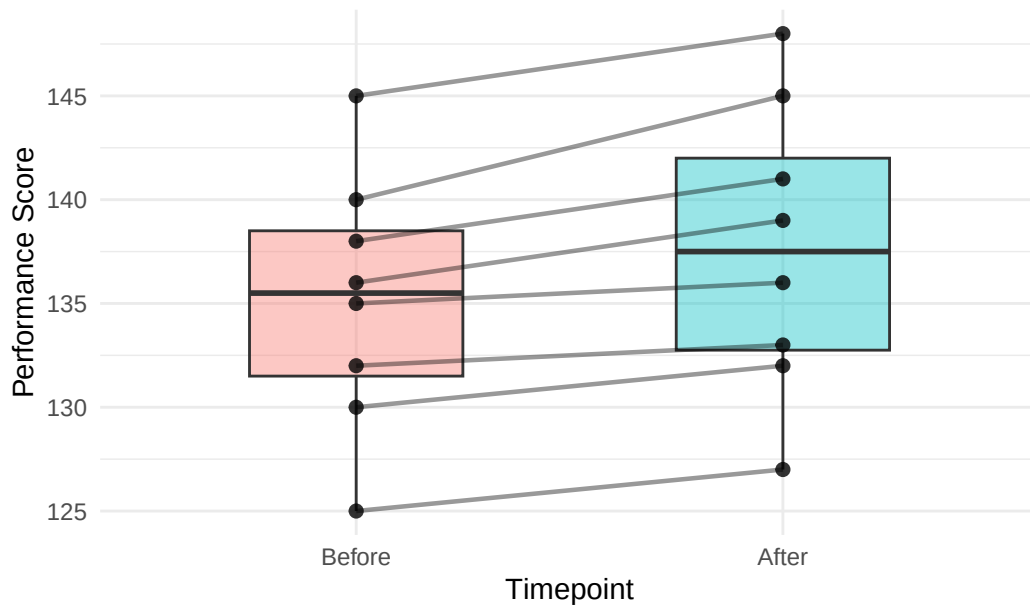
```
data: data_wide$After and data_wide$Before
t = 5.4006, df = 7, p-value = 0.001008
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 1.405392 3.594608
sample estimates:
mean difference
      2.5
```

A paired-samples t-test was performed to determine whether the cardiac rehabilitation training had a significant effect on participant performance over time. Performance scores significantly increased following the intervention ($t(7)=5.40$, $p=0.001$), with a mean improvement of 2.50 points (95% CI [1.41,3.59]). Because $p<0.05$, we reject the null hypothesis and conclude that the rehabilitation program had a statistically significant positive effect on performance.

Visualizing the data

```
ggplot(data_long, aes(x = time, y = perf, fill = time)) +
  geom_boxplot(alpha = 0.4, outlier.shape = NA, width = 0.5) +
  geom_point(aes(group = ID), alpha = 0.8, size = 2) +
  geom_line(aes(group = ID), alpha = 0.4, linewidth = 0.8) +
  labs(
    title = "Performance Scores Before and After Cardiac Rehabilitation",
    x = "Timepoint",
    y = "Performance Score"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

Performance Scores Before and After Cardiac Rehabilitation



Individual and aggregate changes in performance scores before and after cardiac rehabilitation. The boxplots represent the group's median and interquartile ranges at both timepoints. The connecting lines trace individual patient trajectories, illustrating that every single participant experienced an increase in their performance score following the intervention. A paired t-test confirmed this overall improvement (Mean difference = +2.5) is statistically significant ($t = 5.40$, $p = 0.001$).

Questions

- Does the treatment have an effect?
 - Yes. The physical performance scores increased for all 8 patients following the cardiac rehabilitation program. According to the paired test, the the rehabilitation program had a statistically significant positive effect ($t(7)=5.40$, $p=0.001$) on performance. The mean improvement was 2.50 points.
- What comparison are you going to make?
 - Which test will you use? Parametric or non-parametric?
 - * The parametric paired t-test. The Shapiro-Wilk test on the score differences yielded a p-value of 0.2828. Since this is greater than 0.05, the data meets the required assumption of normality, allowing for a parametric test.
 - Which graph illustrates the effect of the treatment?

- * A paired boxplot with connecting lines tracing individual patient trajectories. This design visualizes both the aggregate group shift and the distinct response of each patient from before to after the treatment.
- What sentence will you write in your thesis to explain your result?
- * Following the cardiac rehabilitation training, patients demonstrated a statistically significant increase in physical performance scores, improving by an average of 2.5 units (paired t-test: $t = 5.40$, $df = 7$, $p = 0.001$).

2.3 Testing some stereotypes

```
library(tidyverse)
```

Load the data

```
# Load the snoring dataset from the Data folder.
snore <- read.table("Data/snore.txt", header = TRUE)

# Check the structure of the dataset.
str(snore)
```

```
'data.frame': 100 obs. of 7 variables:
 $ age : int 47 56 46 70 51 46 40 46 49 39 ...
 $ weight: int 71 58 116 96 91 98 112 77 76 119 ...
 $ height: int 158 164 208 186 195 188 193 165 164 196 ...
 $ alcool: int 0 7 3 3 2 0 5 0 0 3 ...
 $ sex : chr "H" "H" "H" "H" ...
 $ snore : chr "N" "O" "N" "N" ...
 $ tabac : chr "O" "N" "O" "O" ...
```

```
# Display the first observations.
head(snore)
```

	age	weight	height	alcool	sex	snore	tabac
1	47	71	158	0	H	N	O
2	56	58	164	7	H	O	N
3	46	116	208	3	H	N	O

4	70	96	186	3	H	N	0
5	51	91	195	2	H	0	0
6	46	98	188	0	F	N	N

Check the categorical variables

```
# Number of participants
nrow(snore)
```

```
[1] 100
```

```
# Number of participants according to snoring status
table(snore$snore)
```

```
  N  0
65 35
```

```
# Number of participants according to sex
table(snore$sex)
```

```
  F  H
25 75
```

```
# Number of participants according to smoking status
table(snore$tabac)
```

```
  N  0
36 64
```

Are snorers fatter?

To compare body size between snorers and non-snorers, we calculate the Body Mass Index (BMI) from body weight and height, which accounts for differences in body size.

```
# Calculate height in metres
snore$height_m <- snore$height / 100

# Calculate Body Mass Index (BMI)
snore$BMI <- snore$weight / (snore$height_m^2)

# Check the calculated BMI values
head(snore[, c("weight", "height", "BMI")])
```

	weight	height	BMI
1	71	158	28.44095
2	58	164	21.56454
3	116	208	26.81213
4	96	186	27.74887
5	91	195	23.93162
6	98	188	27.72748

We calculate the mean and median BMI in each snoring group to describe and compare the central tendency of the two groups.

```
# Mean BMI for each snoring group
tapply(snore$BMI, snore$snore, mean)
```

	N	O
	27.27166	27.11372

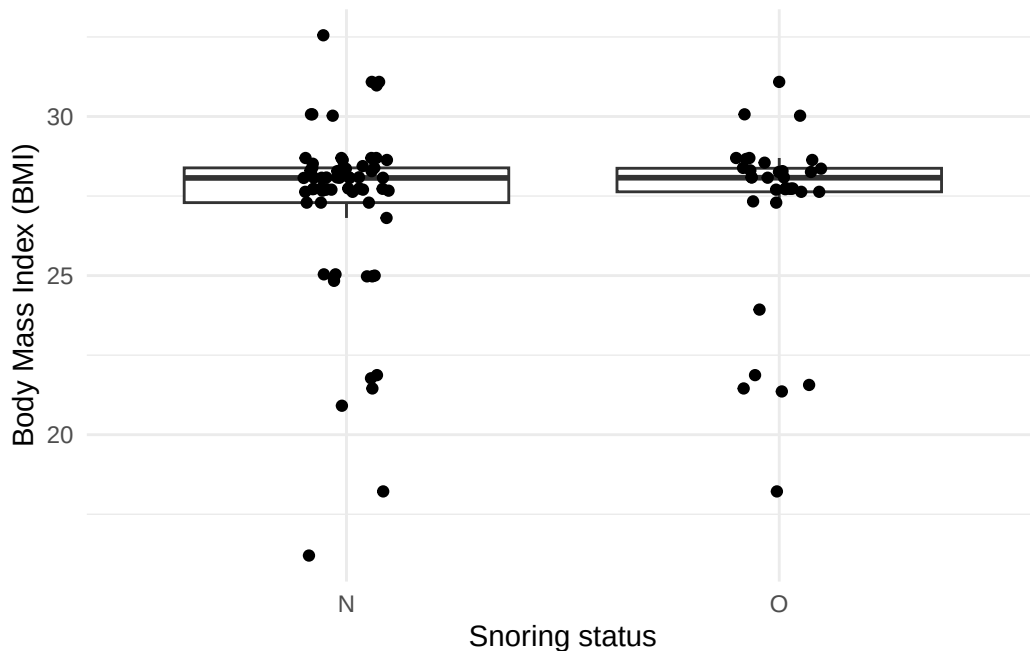
```
# Median BMI for each snoring group
tapply(snore$BMI, snore$snore, median)
```

	N	O
	28.06892	28.07632

We visualize the BMI distributions in snorers and non-snorers to assess their variability and identify possible differences between the two groups.

```
# Compare BMI distributions between snorers and non-snorers
ggplot(snore, aes(x = snore, y = BMI)) +
  geom_boxplot(outlier.shape = NA) +
  geom_jitter(width = 0.1) +
```

```
labs(
  x = "Snoring status",
  y = "Body Mass Index (BMI)"
)
```



We use a Mann–Whitney U test to determine whether BMI differs significantly between snorers and non-snorers.

```
# Perform a Mann-Whitney U test
# H0: BMI does not differ between snorers and non-snorers
# H1: BMI differs between snorers and non-snorers

wilcox.test(BMI ~ snore, data = snore)
```

Wilcoxon rank sum test with continuity correction

data: BMI by snore

W = 1128.5, p-value = 0.951

alternative hypothesis: true location shift is not equal to 0

The p-value is compared with the significance level of 0.05 to determine whether the observed difference in BMI between the two groups is statistically significant.

Results :

The mean BMI was 27.27 kg/m² in non-snorers and 27.11 kg/m² in snorers, while the median BMI was 28.07 and 28.08 kg/m², respectively. The Mann-Whitney U test showed no statistically significant difference in BMI between the two groups ($W = 1128.5$, $p = 0.951$). Therefore, these data do not provide evidence that snorers have a different BMI from non-snorers.

Do snorers drink more ?

We first calculate the mean and median alcohol consumption in each snoring group to describe and compare their alcohol consumption.

```
# Mean alcohol consumption for each snoring group
tapply(snore$alcohol, snore$snore, mean)
```

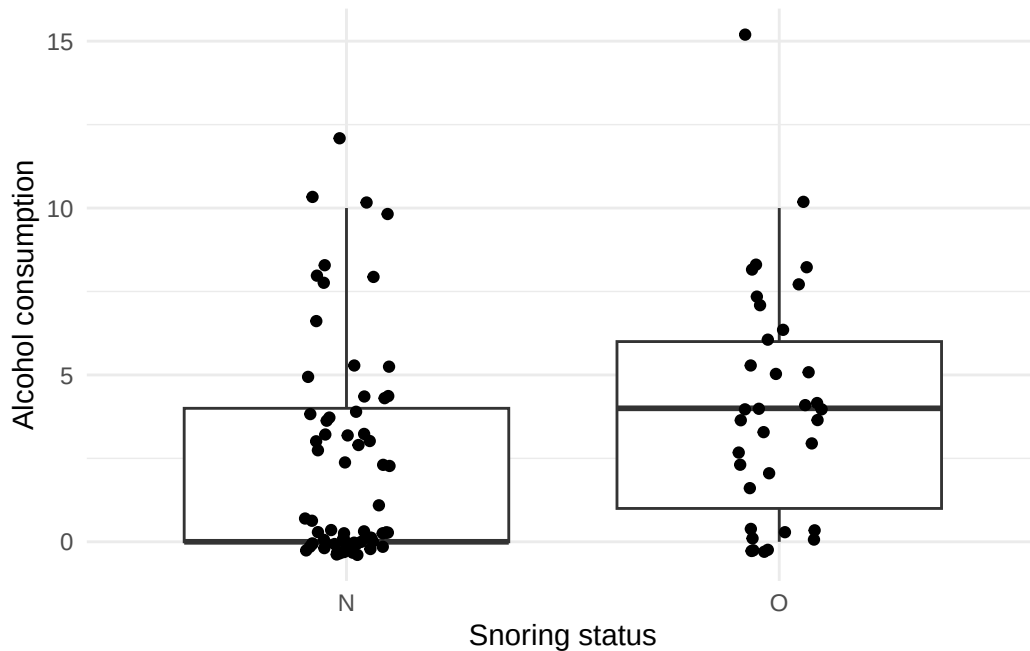
```
      N      0
2.369231 4.028571
```

```
# Median alcohol consumption for each snoring group
tapply(snore$alcohol, snore$snore, median)
```

```
 N 0
 0 4
```

We visualize alcohol consumption in snorers and non-snorers to compare the distributions and identify possible differences or extreme values.

```
# Compare alcohol consumption between snorers and non-snorers
ggplot(snore, aes(x = snore, y = alcohol)) +
  geom_boxplot(outlier.shape = NA) +
  geom_jitter(width = 0.1) +
  labs(
    x = "Snoring status",
    y = "Alcohol consumption"
  )
```



We use a Mann–Whitney U test to determine whether alcohol consumption differs significantly between snorers and non-snorers.

```
# Perform a Mann-Whitney U test
# H0: alcohol consumption does not differ between snorers and non-snorers
# H1: alcohol consumption differs between snorers and non-snorers

wilcox.test(alcool ~ snore, data = snore)
```

Wilcoxon rank sum test with continuity correction

```
data: alcool by snore
W = 788, p-value = 0.008594
alternative hypothesis: true location shift is not equal to 0
```

Results :

Alcohol consumption was higher among snorers (mean = 4.03, median = 4) than among non-snorers (mean = 2.37, median = 0). For the statistical test part, since $p = 0.008594 < 0.05$, we reject the null-hypothesis H_0 . A statistically significant difference in alcohol consumption was observed between snorers and non-snorers (Mann–Whitney U test, $W = 788$, $p = 0.0086$).

Do snorers smoke more ?

We calculate the number and proportion of smokers in each snoring group to determine whether smoking is more frequent among snorers.

```
# Number of smokers and non-smokers in each snoring group
table(snore$snore, snore$tabac)
```

```
      N  0
N 21 44
O 15 20
```

```
# Proportion of smokers within each snoring group
prop.table(table(snore$snore, snore$tabac), margin = 1)
```

```
      N      0
N 0.3230769 0.6769231
O 0.4285714 0.5714286
```

We visualize the proportion of smokers in each snoring group to make the difference between the two groups easier to interpret.

```
# Create a contingency table
smoking_table <- prop.table(
  table(snore$snore, snore$tabac),
  margin = 1
)

# Convert the table to a data frame
smoking_data <- as.data.frame(smoking_table)

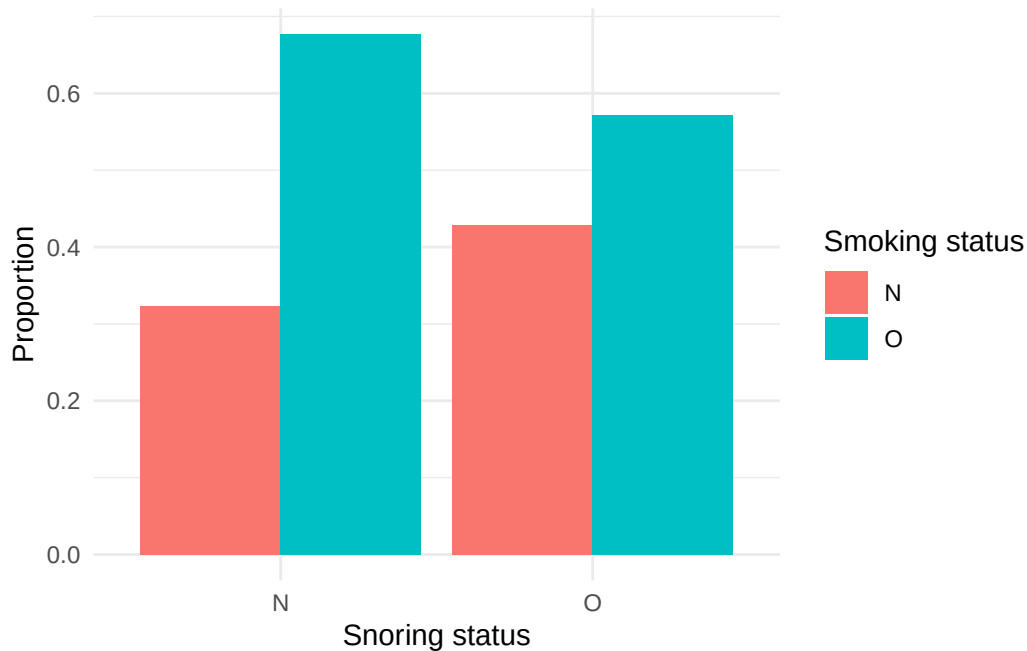
# Rename the columns
colnames(smoking_data) <- c("Snoring", "Smoking", "Proportion")

# Plot the proportions
ggplot(smoking_data, aes(x = Snoring, y = Proportion, fill = Smoking)) +
  geom_col(position = "dodge") +
  labs(
    x = "Snoring status",
```

```

y = "Proportion",
fill = "Smoking status"
)

```



We use Fisher's exact test to determine whether smoking status is statistically associated with snoring status. Fisher's exact test was used because both snoring status and smoking status are categorical variables forming a 2 * 2 contingency table.

```

# Create a contingency table
smoking_table <- table(snore$snore, snore$tabac)

# Perform Fisher's exact test
# H0: smoking status and snoring status are independent
# H1: smoking status and snoring status are associated

fisher.test(smoking_table)

```

Fisher's Exact Test for Count Data

```

data: smoking_table
p-value = 0.3827

```

```

alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 0.2516104 1.6282374
sample estimates:
odds ratio
 0.6393445

```

Results :

The proportion of smokers was 67.7% among non-snorers and 57.1% among snorers. Fisher's exact test showed no statistically significant association between smoking status and snoring status ($p = 0.3827$).

Are men fatter ?

We calculate the mean and median BMI for men and women to describe and compare body mass index between the two sex groups.

```

# Mean BMI for each sex group
tapply(snore$BMI, snore$sex, mean)

```

```

      F      H
27.04192 27.27453

```

```

# Median BMI for each sex group
tapply(snore$BMI, snore$sex, median)

```

```

      F      H
27.72748 28.07504

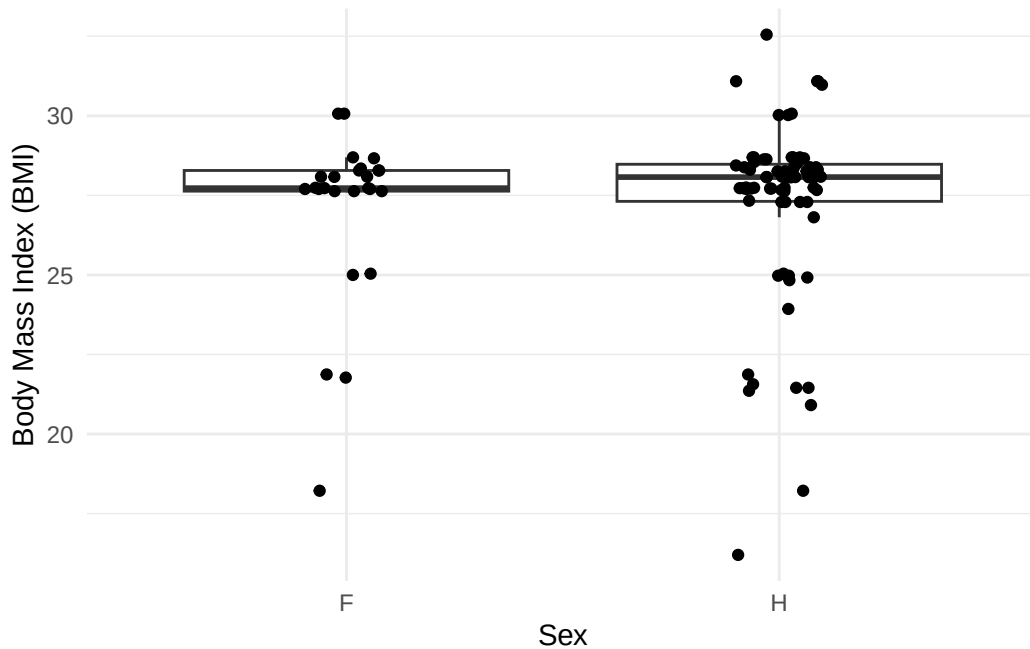
```

We visualize the BMI distributions of men and women to assess their variability and identify possible differences between the two groups.

```

# Compare BMI distributions between men and women
ggplot(snore, aes(x = sex, y = BMI)) +
  geom_boxplot(outlier.shape = NA) +
  geom_jitter(width = 0.1) +
  labs(
    x = "Sex",
    y = "Body Mass Index (BMI)"
  )

```



We use a Mann–Whitney U test to determine whether BMI differs significantly between men and women.

```
# Perform a Mann-Whitney U test
# H0: BMI does not differ between men and women
# H1: BMI differs between men and women

wilcox.test(BMI ~ sex, data = snore)
```

Wilcoxon rank sum test with continuity correction

```
data: BMI by sex
W = 850, p-value = 0.4884
alternative hypothesis: true location shift is not equal to 0
```

Results :

The mean BMI was 27.04 kg/m² in women and 27.27 kg/m² in men, while the median BMI was 27.73 and 28.08 kg/m², respectively. No statistically significant difference in BMI was observed between women and men (Mann–Whitney U test, $W = 850$, $p = 0.4884$).

Do women smoke less?

We calculate the number and proportion of smokers among men and women to determine whether smoking is less frequent among women.

```
# Number of smokers and non-smokers for each sex
table(snore$sex, snore$tabac)
```

```
      N  0
F 15 10
H 21 54
```

```
# Proportion of smokers within each sex group
prop.table(table(snore$sex, snore$tabac), margin = 1)
```

```
      N    0
F 0.60 0.40
H 0.28 0.72
```

We visualize the proportion of smokers in each sex group to facilitate comparison between men and women.

```
# Create a contingency table
sex_smoking_table <- prop.table(
  table(snore$sex, snore$tabac),
  margin = 1
)

# Convert the table to a data frame
sex_smoking_data <- as.data.frame(sex_smoking_table)

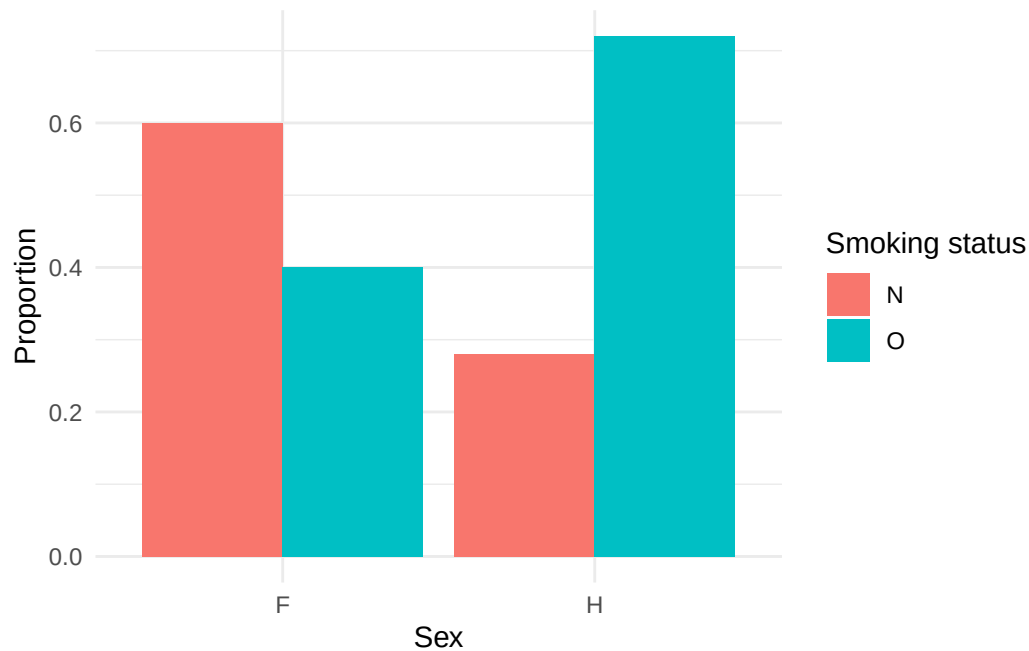
# Rename the columns
colnames(sex_smoking_data) <- c("Sex", "Smoking", "Proportion")

# Plot the proportions
ggplot(sex_smoking_data, aes(x = Sex, y = Proportion, fill = Smoking)) +
  geom_col(position = "dodge") +
  labs(
    x = "Sex",
```

```

y = "Proportion",
fill = "Smoking status"
)

```



We use Fisher's exact test to determine whether smoking status is statistically associated with sex.

```

# Create a contingency table
sex_smoking_table <- table(snore$sex, snore$tabac)

# Perform Fisher's exact test
# H0: sex and smoking status are independent
# H1: sex and smoking status are associated

fisher.test(sex_smoking_table)

```

Fisher's Exact Test for Count Data

```

data: sex_smoking_table
p-value = 0.007276
alternative hypothesis: true odds ratio is not equal to 1

```

```
95 percent confidence interval:
  1.354726 11.139908
sample estimates:
odds ratio
  3.79892
```

Results :

Smoking was reported by 40% of women and 72% of men. Fisher's exact test showed a statistically significant association between sex and smoking status ($p = 0.0073$).

From a more general perspective : are there any correlation between variables?

We select the quantitative variables to investigate possible associations between age, body measurements, alcohol consumption, and BMI.

```
# Select the quantitative variables
quantitative_data <- snore[, c(
  "age",
  "weight",
  "height",
  "alcohol",
  "BMI"
)]

# Check the selected data
head(quantitative_data)
```

	age	weight	height	alcohol	BMI
1	47	71	158	0	28.44095
2	56	58	164	7	21.56454
3	46	116	208	3	26.81213
4	70	96	186	3	27.74887
5	51	91	195	2	23.93162
6	46	98	188	0	27.72748

We calculate a Spearman correlation matrix to identify monotonic associations between the quantitative variables.

```
# Calculate the Spearman correlation matrix
cor(
  quantitative_data,
  method = "spearman"
)
```

	age	weight	height	alcohol	BMI
age	1.000000000	-0.01455125	0.008072461	-0.10748796	-0.04801977
weight	-0.014551254	1.000000000	0.930383489	0.08568687	0.54367358
height	0.008072461	0.93038349	1.000000000	0.12539769	0.29991626
alcohol	-0.107487956	0.08568687	0.125397694	1.000000000	0.02115004
BMI	-0.048019774	0.54367358	0.299916261	0.02115004	1.000000000

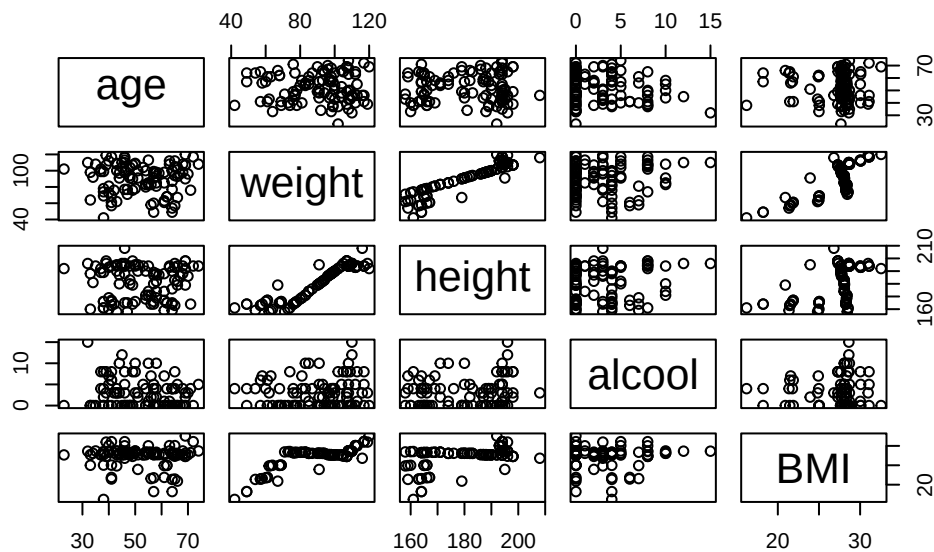
A very strong positive correlation was observed between weight and height (Spearman's $\rho = 0.930$). A moderate positive correlation was observed between weight and BMI ($\rho = 0.544$), while the correlation between height and BMI was weaker ($\rho = 0.300$). The other associations were weak, including those between age and BMI ($\rho = -0.048$) and between alcohol consumption and BMI ($\rho = 0.021$).

Because BMI is calculated from weight and height, correlations between BMI and these variables are partly mathematical and should therefore be interpreted with caution.

Visualization with pairwise scatterplots

We visualize the relationships between the quantitative variables using pairwise scatterplots to identify possible linear or monotonic relationships.

```
# Create pairwise scatterplots for the quantitative variables
pairs(quantitative_data)
```



The scatterplots show a strong positive relationship between weight and height, while the other relationships appear weaker. In particular, the weight–height relationship is clearly visible as the points follow an increasing trend.

Test the correlations

We perform Spearman correlation tests to determine whether the observed associations between the quantitative variables are statistically significant.

```
# Test the main correlations between quantitative variables

# Weight and height
cor.test(snore$weight, snore$height, method = "spearman")
```

Spearman's rank correlation rho

```
data: snore$weight and snore$height
S = 11602, p-value < 2.2e-16
alternative hypothesis: true rho is not equal to 0
sample estimates:
rho
0.9303835
```

```
# Weight and BMI
cor.test(snore$weight, snore$BMI, method = "spearman")
```

Spearman's rank correlation rho

```
data: snore$weight and snore$BMI
S = 76047, p-value = 5.056e-09
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.5436736
```

```
# Height and BMI
cor.test(snore$height, snore$BMI, method = "spearman")
```

Spearman's rank correlation rho

```
data: snore$height and snore$BMI
S = 116669, p-value = 0.002433
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.2999163
```

```
# Age and BMI
cor.test(snore$age, snore$BMI, method = "spearman")
```

Spearman's rank correlation rho

```
data: snore$age and snore$BMI
S = 174652, p-value = 0.6352
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.04801977
```

```
# Alcohol consumption and BMI
cor.test(snore$alcohol, snore$BMI, method = "spearman")
```

Spearman's rank correlation rho

```
data: snore$alcohol and snore$BMI
S = 163125, p-value = 0.8346
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.02115004
```

A strong positive correlation was found between weight and height (Spearman's $\rho = 0.930$, $p < 2.2e-16$). Moderate and weak positive correlations were also observed between weight and BMI ($p = 0.544$, $p < 0.001$) and between height and BMI ($p = 0.300$, $p = 0.0024$), respectively. No statistically significant correlation was found between age and BMI ($p = -0.048$, $p = 0.635$) or between alcohol consumption and BMI ($p = 0.021$, $p = 0.835$).

Results

Overall, the results provided mixed evidence for the tested stereotypes, with significant differences found for alcohol consumption among snorers and smoking according to sex. The correlation analysis mainly highlighted strong associations between anthropometric variables, particularly weight and height.

Epilogue: beyond comparing groups

The tests above answer a descriptive question: *is there a difference between groups?* Data mining asks a different one: *can we use what we measured to predict or to group?* With the snoring data, a logistic regression could estimate the probability that a person snores from their age, BMI, alcohol and tobacco consumption, and tell us which factors contribute once the others are taken into account, something that separate two-group tests cannot do. Methods such as PCA or clustering go further and look for profiles of individuals without any outcome chosen in advance. These tools are only useful if we know why we use them: a prediction model needs a clear question, enough participants for the number of predictors, and validation on new data, otherwise it only describes the noise of one sample. In clinical or sport research, the value lies less in the algorithm than in the decision it supports, such as identifying patients who would benefit most from rehabilitation.

Our Git workflow

Principle. Git produces merge conflicts when two people edit the same lines in parallel. We therefore split the report into one master document (`IEAP-Series02-RStudio.qmd`) and several sub-documents (`sections/_*.qmd`) that are included at render time.

Branches.

- `main` is the graded branch. Nobody commits directly on it (except for the initial commit); it receives merges through pull requests on GitHub.
- One feature branch per question, created from an up-to-date `main`: `2.1-concepts` (Alexis), `2.2-prepost` (Vidusha), `2.3-Testing-some-sterotypes` (Chloé), `3-epilogue` (Alexis).
- `git-checklist` for the shared section.
- `final`: last branch, used only to render and commit the PDF once everything is merged.

Cycle for each question.

1. Create a new branch for a new section
2. Work, then commit small, meaningful steps, e.g.
3. Push the branch and create a pull request on GitHub
4. Another team member reads the pull request and approves it; we merge into `main`

Commit messages. Each message contains what was added or changed with the commit

Challenges encountered

- **Alexis:** Had to figure it out how the section folder is working, so we can work on the same project without overlapping what the group is doing. How to connect a reference bibliography to `.qmd`.
- **Vidusha:** I often forget to commit after each change, this assignment reminds me to commit a significant change.
- **Chloé:** My branch was one commit behind because i cloned it before a change was made on main branch. I had to figure it out how to have my branch up to date. My R cells weren't computing, after restarting R environment, all worked together!

Checklist

Repository

- ☐ Public GitHub repository named `IEAP-Series02-RStudio`
- ☐ `main` is up to date (all pull requests merged)

- ☐ README.md explains the project, the structure, the team and how to render
- ☐ LICENSE file present
- ☐ IEAP-Series02-RStudio.qmd present and renders without error
- ☐ IEAP-Series02-RStudio.pdf committed on main and uploaded to Moodle

Git history

- ☐ Meaningful commit messages, prefixed with the question number
- ☐ At least one commit per question
- ☐ One branch per question, merged into main through pull requests
- ☐ No direct commit on main except the initial commit
- ☐ Every team member has commits in the history

Report content

- ☐ Authors and date in the header
- ☐ Clickable link to the public GitHub repository
- ☐ Analysis of our Git workflow
- ☐ Challenges and lessons learned
- ☐ All questions of 2.1 answered, each with at least one reference
- ☐ 2.2: variable table, comparison, test choice, figure, result sentence
- ☐ 2.3: data, figure and result sentence for each of the five questions
- ☐ Complete code visible in the PDF
- ☐ Data sources with their URLs
- ☐ Scientific references with their DOIs

Data sources

File	Description	Source (URL)
data/PrePost.csv	Before/after measurements, cardiac rehabilitation	https://moodle.umontpellier.fr/pluginfile.php/20250923.zip?forcedownload=1
data/snore.txt	Anthropometric and qualitative data, 1 person per line	https://moodle.umontpellier.fr/pluginfile.php/20250923.zip?forcedownload=1

Software

Analyses were run in R (R version 4.5.1 (2025-06-13 ucrt)) with the tidyverse ([wickham2019?](#)), and the report was written with Quarto.

References

- Altman, Douglas G., and J. Martin Bland. 1995. "Statistics Notes: The Normal Distribution." *BMJ* 310 (6975): 298. <https://doi.org/10.1136/bmj.310.6975.298>.
- . 2005. "Standard Deviations and Standard Errors." *BMJ* 331 (7521): 903. <https://doi.org/10.1136/bmj.331.7521.903>.
- Banerjee, Amitav, U. B. Chitnis, S. L. Jadhav, J. S. Bhawalkar, and S. Chaudhury. 2009. "Hypothesis Testing, Type I and Type II Errors." *Industrial Psychiatry Journal* 18 (2): 127–31. <https://doi.org/10.4103/0972-6748.62274>.
- Bland, J. Martin, and Douglas G. Altman. 2009. "Analysis of Continuous Data from Small Samples." *BMJ* 338: a3166. <https://doi.org/10.1136/bmj.a3166>.
- Ghasemi, Asghar, and Saleh Zahediasl. 2012. "Normality Tests for Statistical Analysis: A Guide for Non-Statisticians." *International Journal of Endocrinology and Metabolism* 10 (2): 486–89. <https://doi.org/10.5812/ijem.3505>.
- Greenland, Sander, Stephen J. Senn, Kenneth J. Rothman, John B. Carlin, Charles Poole, Steven N. Goodman, and Douglas G. Altman. 2016. "Statistical Tests, p Values, Confidence Intervals, and Power: A Guide to Misinterpretations." *European Journal of Epidemiology* 31 (4): 337–50. <https://doi.org/10.1007/s10654-016-0149-3>.
- Manikandan, S. 2011. "Measures of Central Tendency: Median and Mode." *Journal of Pharmacology and Pharmacotherapeutics* 2 (3): 214–15. <https://doi.org/10.4103/0976-500X.83300>.
- Nahm, Francis Sahngun. 2016. "Nonparametric Statistical Tests for the Continuous Data: The Basic Concept and the Practical Use." *Korean Journal of Anesthesiology* 69 (1): 8–14. <https://doi.org/10.4097/kjae.2016.69.1.8>.
- Wasserstein, Ronald L., and Nicole A. Lazar. 2016. "The ASA Statement on p-Values: Context, Process, and Purpose." *The American Statistician* 70 (2): 129–33. <https://doi.org/10.1080/00031305.2016.1154108>.
- Welch, B. L. 1947. "The Generalization of 'Student's' Problem When Several Different Population Variances Are Involved." *Biometrika* 34 (1–2): 28–35. <https://doi.org/10.1093/biomet/34.1-2.28>.
- Wilcoxon, Frank. 1945. "Individual Comparisons by Ranking Methods." *Biometrics Bulletin* 1 (6): 80–83. <https://doi.org/10.2307/3001968>.